Vector spaces

vector space a set V with field F and operations $+$ and $\cdot$ such that $\forall u, v, w \in V, \forall a, b \in F$:

- $u + v \in V$
- $a \cdot u \in V$
- $u + v = v + u$
- $(u + v) + w = u + (v + w)$
- $\exists 0 \in V : u + 0 = u$
- $\exists (-u) \in V : u + (-u) = 0$
- $1 \cdot u = u$
- $(ab) \cdot u = a \cdot (b \cdot u)$
- $(a + b) \cdot u = a \cdot u + b \cdot u$
- $a \cdot (u + v) = a \cdot u + a \cdot v$

subspace a subset W of V that is itself a vector space under the same operations. It is sufficient that $W \neq \varnothing$ and $\forall u, v \in W \forall a, b \in F, au + bv \in W$.

linear combination for a subset S of V, a sum of scalar multiples of vectors in S, $\sum_{v \in S} a_v v$

linear independence the property of a subset S of V where $\sum_{v \in S} a_v v = 0 \implies \forall v \in S, a_v = 0$

span $\text{Span}(S) := \left\{ \sum_{v \in S} a_v v \,\middle|\, \forall s \in S, a_s \in F \right\}$, the set of all linear combinations of vectors in S.